

Shivesh Saravanan

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PROFESSIONAL SUMMARY

AI/ML Engineer with **2+ years** of production experience architecting, fine-tuning, and deploying **Generative AI, LLMs**, and enterprise-scale **MLOps pipelines** on **AWS** and **OCI**. Proven track record in **Fine-Tuning (QLoRA, DPO)**, **Multi-Agent RAG**, **Distributed Training**, and **Inference Optimization (vLLM, TensorRT-LLM)**. Specialized in building sub-80ms model serving infrastructures and scalable telemetry pipelines that eliminated 14,000 downtime hours.

TECHNICAL SKILLS

Languages & Core Frameworks: Python, SQL, C++, Rust, Java, Go, Bash, PyTorch, JAX, TensorFlow, Scikit-Learn, Keras, Hugging Face (Transformers, Accelerate), OpenCV, NumPy, Pandas, FastAPI, Flask, gRPC, REST APIs

Generative AI & Agentic Systems: LLMs, Transformers, Fine-Tuning, RAG, Agentic Workflows, LangChain, LlamaIndex, LangGraph, AutoGen, DSPy, Synthetic Data Generation, Prompt Engineering, RLHF, DPO, KTO

Inference, Guardrails & Eval: vLLM, SGLang, Triton Inference Server, TensorRT-LLM, FlashAttention, Quantization (AWQ, GPTQ, FP8), NeMo Guardrails, Ragas, DeepEval, ONNX Runtime, CUDA Optimization

MLOps & Distributed AI: DeepSpeed, Ray (Train/Serve), Unsloth, Weights & Biases (W&B), MLflow, Docker, Kubernetes (EKS), CI/CD, Feature Stores, Model & Data Drift Monitoring

Cloud & Big Data Infrastructure: AWS (SageMaker, S3, EC2, Lambda), Oracle Cloud Infrastructure (OCI), Databricks, Snowflake, Apache Spark, Hadoop, Kafka, Airflow, Distributed Computing

Databases & Vector Search: Vector Databases (Pinecone, FAISS), PostgreSQL, MongoDB, Redis, Elasticsearch

EXPERIENCE

Dell Technologies

Associate AI/ML Engineer

India

Jul 2022 – Aug 2024

- Architected a distributed **predictive failure detection** pipeline using **Ray**, **Apache Spark**, and **PyTorch** on **AWS SageMaker**, analyzing 18 TB monthly telemetry to eliminate 14,000 downtime hours and save **\$1M**.
- Fine-tuned **LLaMA-3** and **BERT** models via **PEFT/QLoRA** and **LangChain** for agentic IT ticket routing, slashing **Mean-Time-To-Resolution** from 48 to 11 hours while saving 6,200 manual triage hours annually.
- Engineered real-time **LLM model serving** infrastructure using **vLLM**, **TensorRT-LLM**, and **Triton** on **AWS SageMaker** with **AWQ quantization**, sustaining **sub-80ms latency** across 12K requests/hour with zero downtime.
- Deployed a **Computer Vision** defect-detection system using **YOLOv8** and **PyTorch** optimized with **TensorRT**, achieving a **0.97 F1-score** for hardware QA while saving **\$1.8M** annually in warranty rework.
- Scaled feature engineering workflows on **Databricks** and **Apache Spark** using **Feast Feature Store** and **Pinecone vector store**, cutting data pipeline latency 4.2x and accelerating retraining cycles by 3 weeks.
- Built automated **MLOps CI/CD pipelines** using **MLflow**, **Kubernetes (EKS)**, and **GitHub Actions** for continuous **model versioning**, **A/B testing**, and **data drift monitoring**, accelerating release cadences 6x over prior schedules.

PROJECTS

Multi-Agent Enterprise RAG & LLM Document Synthesizer

- Engineered an agentic **RAG pipeline** using **LangChain**, **Pinecone**, and **HyDE** to query enterprise documents, integrating **NeMo Guardrails** for hallucination mitigation and context-aware security defenses.
- Deployed **LLaMA-3 70B** inference serving via **vLLM** and **AWQ 4-bit quantization** on AWS GPU clusters, achieving **3.5x throughput acceleration** and **sub-80ms time-to-first-token** across extensive evaluations.

Distributed LLM Fine-Tuning & Quantization Engine

- Designed a multi-GPU **distributed training framework** using **PyTorch**, **DeepSpeed ZeRO-3**, and **Ray Train** to fine-tune open-source LLMs via **QLoRA** and **Direct Preference Optimization** methods.
- Compiled fine-tuned models with **TensorRT-LLM** and **ONNX Runtime**, achieving a **nearly three-fold reduction** in GPU memory footprint while automating **MLflow** evaluation tracking across multi-node clusters.

EDUCATION

The University of Texas at Arlington

Master of Science in Computer Science

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science and Engineering

Arlington, TX, USA

Aug 2024 – May 2026

India

Sep 2020 – Jul 2024

CERTIFICATIONS

- Oracle Cloud Infrastructure (OCI) Data Foundations Associate** – Oracle
- Introduction to Big Data with Spark and Hadoop** – IBM (Coursera)